

Pre lab 4 activity 1

```
yahboom@yahboom-virtual-machine: ~/intro_ws
INFO [1773430145.431785131] [square_client]: Goal succeeded! Result: turtle_in
erfaces.msg.TurtlePose(heading=0.0, position=array([-9.1848509e-15, 7.1054274e
15]), dtype=float32))
yahboom@yahboom-virtual-machine: ~/intro_ws$ ros2 run turtle_test make_square --r
s-args --param square_size:=200.0
INFO [1773430217.535322137] [square_client]: Waiting for action server...
INFO [1773430219.792755377] [square_client]: Sending goal request with square_
size=200.0
INFO [1773430219.854117509] [square_client]: Goal accepted
INFO [1773430221.006794745] [square_client]: Received feedback: turtle_interfa
ces.msg.TurtlePose(heading=90.0, position=array([ 2.0000000e+02, -4.1880443e-14]
, dtype=float32))
INFO [1773430223.233584665] [square_client]: Received feedback: turtle_interfa
ces.msg.TurtlePose(heading=180.0, position=array([200., 200.], dtype=float32))
INFO [1773430225.467810467] [square_client]: Received feedback: turtle_interfa
ces.msg.TurtlePose(heading=270.0, position=array([5.684342e-14, 2.000000e+02], d
type=float32))
INFO [1773430227.714687088] [square_client]: Received feedback: turtle_interfa
ces.msg.TurtlePose(heading=0.0, position=array([-2.888186e-14, 5.684342e-14], d
type=float32))
INFO [1773430228.720558058] [square_client]: Goal succeeded! Result: turtle_in
erfaces.msg.TurtlePose(heading=0.0, position=array([-2.888186e-14, 5.684342e-1
], dtype=float32))
yahboom@yahboom-virtual-machine: ~/intro_ws$
re_size

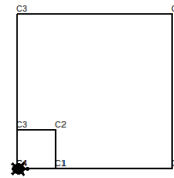
faces/TurtlePose final_pose

faces/TurtlePose current_pose" > action/MakeSquare
om-virtual-machine:~/intro_ws/src/turtle_inter
om-virtual-machine:~/intro_ws/src/turtle_inter
--packages-select turtle_interfaces
ll/setup.bash
ce show turtle_interfaces/action/MakeSquare
turtle_interfaces
turtle_interfaces [18.9s]

ackage finished [19.8s]
re_size

faces/TurtlePose final_pose
t32 heading
t32[2] position

faces/TurtlePose current_pose
t32 heading
t32[2] position
om-virtual-machine:~/intro_ws$ cd ~/intro_ws
ll/setup.bash
tle_test turtle
430138.680510311] [turtle server]: Received goal
```



Pre lab 4 activity 2 unmodified

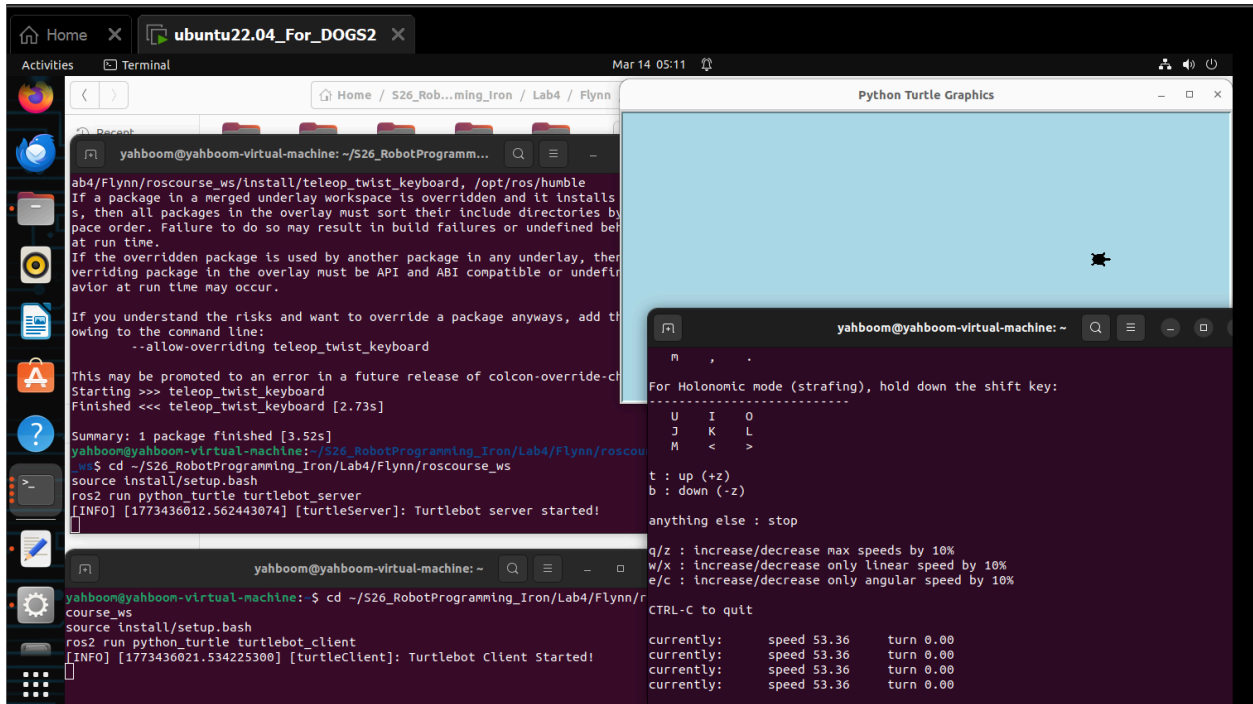
```
yahboom@yahboom-virtual-machine: ~/intro_ws$ ros2 launch python_parameters run_multi_launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-14-03-40-13-398313-yahboom-virtual-machine-7229
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [minimal_param_node-1]: process started with pid [7230]
[INFO] [minimal_param_node-2]: process started with pid [7232]
[INFO] [minimal_param_node-3]: process started with pid [7234]
[minimal_param_node-1] [INFO] [1773430815.137747863] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773430816.107069957] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773430816.634435110] [group0.NA]: Hello default!
[minimal_param_node-1] [INFO] [1773430817.107112954] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773430818.107031467] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773430819.107279585] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773430819.109850204] [group0.NA]: Hello default!
```

Pre lab 4 activity 2 modified

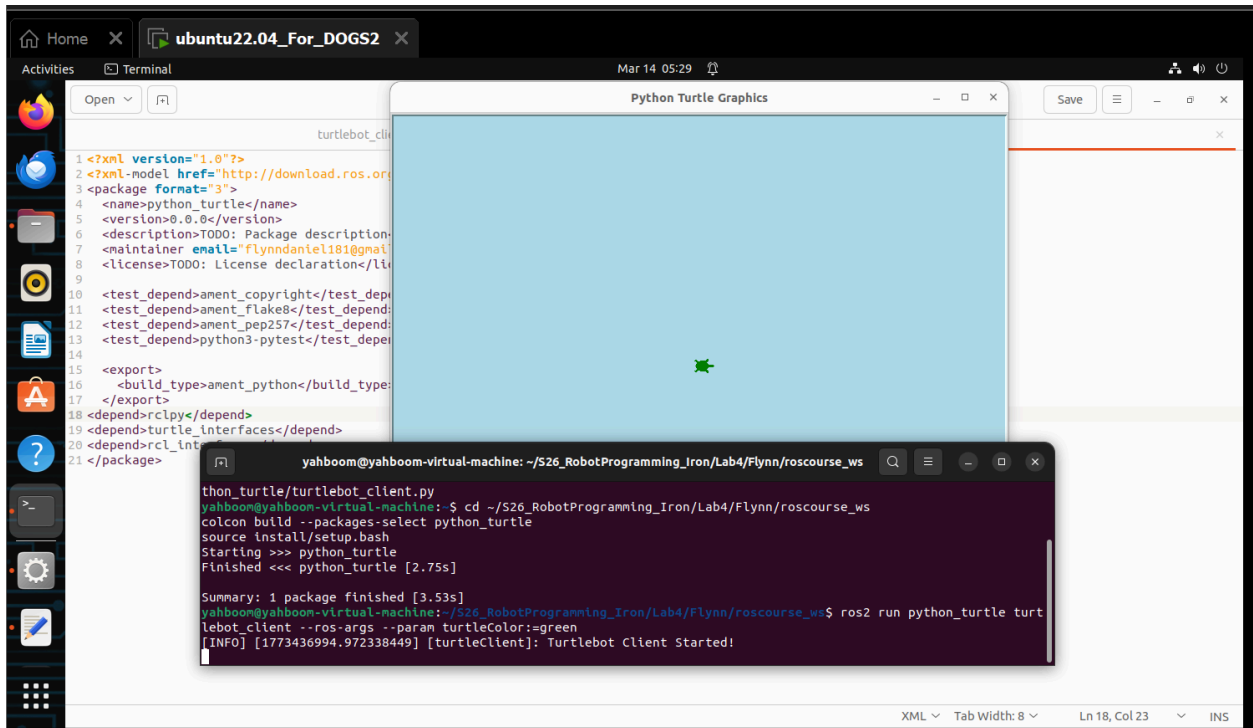
```
yahboom@yahboom-virtual-machine: ~/intro_ws$ ros2 launch python_parameters run_multi_launch.py my_parameter:=World
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-14-03-40-58-788307-yahboom-virtual-machine-7268
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [minimal_param_node-1]: process started with pid [7269]
[INFO] [minimal_param_node-2]: process started with pid [7271]
[INFO] [minimal_param_node-3]: process started with pid [7273]
[minimal_param_node-1] [INFO] [1773430860.756071937] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773430861.732384074] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773430862.016879037] [group0.NA]: Hello World!
[minimal_param_node-1] [INFO] [1773430863.733902867] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773430863.733076679] [group6.Hiro]: Hello Hiro!
[minimal_param_node-3] [INFO] [1773430864.490769398] [group0.NA]: Hello World!
[minimal_param_node-1] [INFO] [1773430864.732791916] [group6.Hiro]: Hello Hiro!
[minimal_param_node-1] [INFO] [1773430865.732483306] [group6.Hiro]: Hello Hiro!
```

Task 1

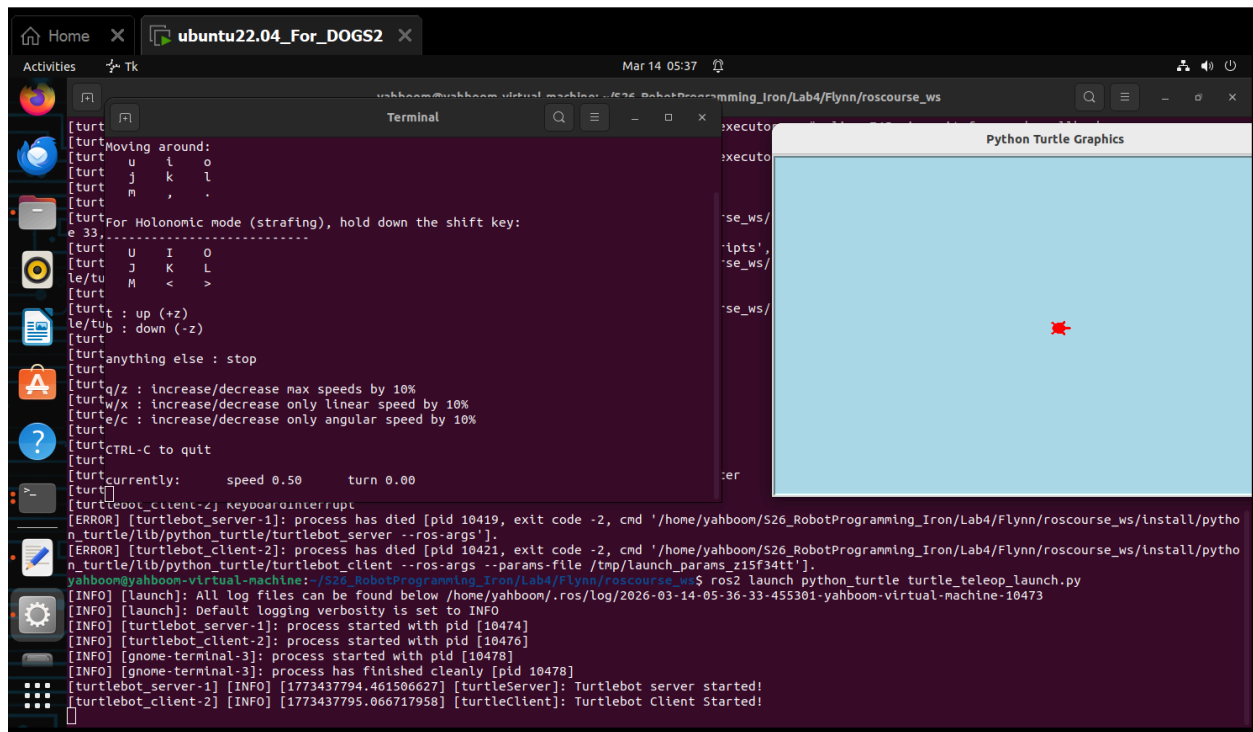
17a. When trying to move the turtle it only slows down when pressing any buttons and spins in a circle to the left.



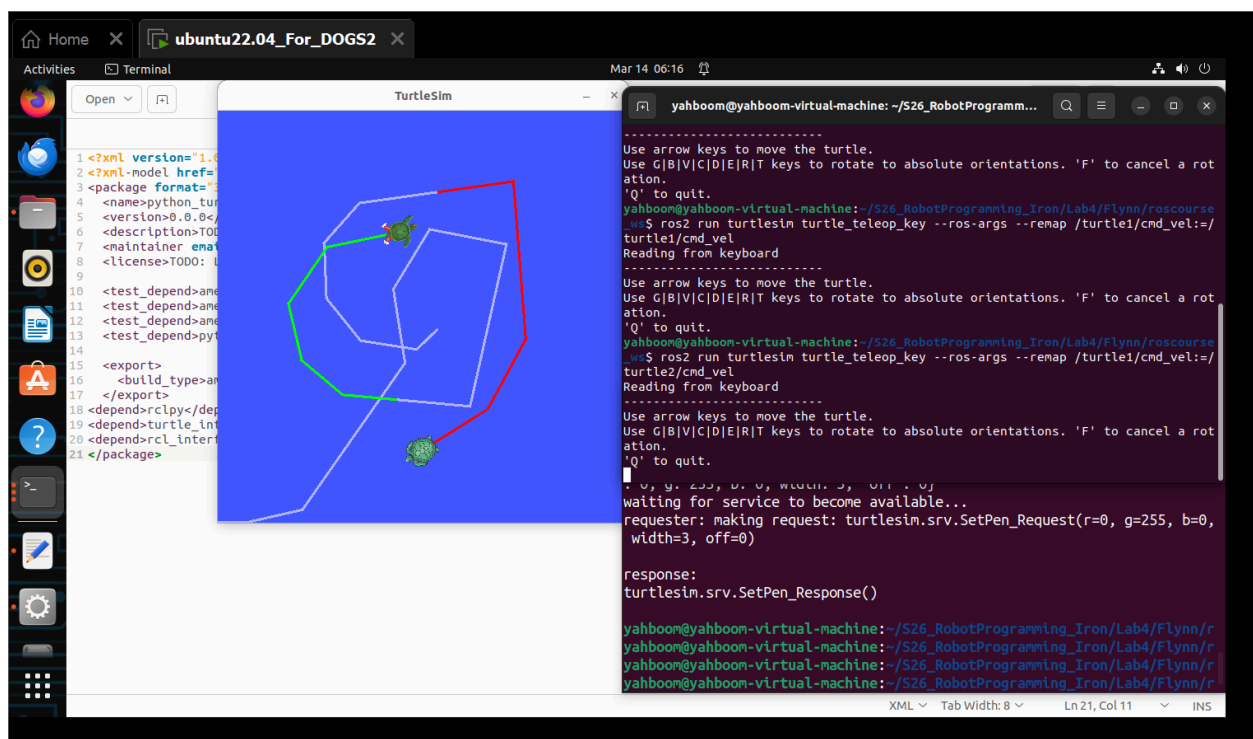
Task 2 step 13



Task 3 step 8



Task 4



Question 5:

The remapping is so useful because it lets you easily switch from controlling in this case one turtle to another and still using the same controls.